ENAE464 - 0202

Lab01: Pressure Measurements of Inviscid Flow over Cylinder

Due on February 13$^{\text{th}}$, 2026 at 11:59 AM

Dr. Silbaugh, 02:00 PM

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Experiment Performed: February 6$^{\text{th}}$, 2026
Report Submitted: February 13$^{\text{th}}$, 2026

February 13th, 2026

Enclosed is the technical report for the Flow Over Cylinder laboratory experiment. This report presents the experimental methodology, results, and analysis of flow characteristics observed around a circular cylinder, including pressure distribution, drag coefficient determination, and comparison with theoretical predictions. Please feel free to contact us with any questions regarding the contents of this report.

Respectfully,
Mikołaj Kostrzewa & Vai Srivastava

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1 Abstract

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2 Introduction

For the inviscid-flow model used in this lab, we assume a potential-flow field that is:

- **Incompressible (continuity):** $\nabla \cdot \vec{\mathbf{V}} = 0$.
- **Irrotational:** $\nabla \times \vec{\mathbf{V}} = 0$.

If $\nabla \times \vec{\mathbf{V}} = 0$, then a *velocity potential* ϕ exists such that $\vec{\mathbf{V}} = \nabla\phi$. Substituting into continuity gives Laplace's equation:

$$\nabla^2\phi = 0 \qquad \left(\text{i.e., } \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2} = 0\right). \tag{2.0.1}$$

In order to model flow over a 2D cylinder, we use an approach presented in [1], which uses a superposition of a uniform flow and a doublet flow, as presented in the figure below [1, Fig. 3.26]:

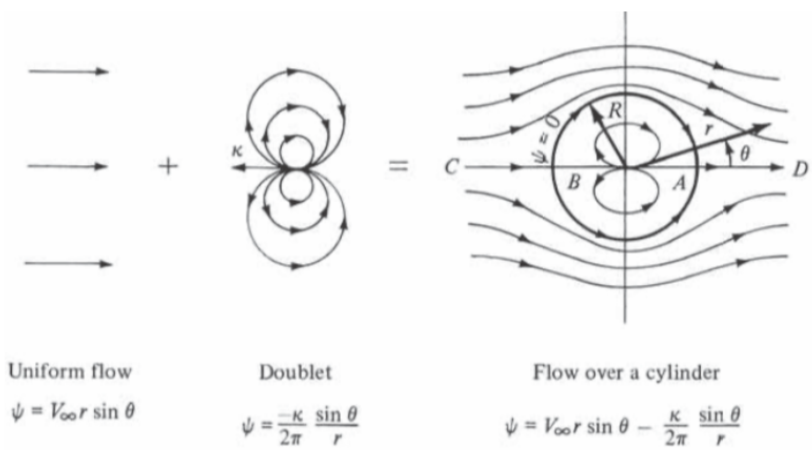

Figure 2.0.1: Superposition of uniform flow and a doublet to model nonlifting potential flow over a circular cylinder.

In polar coordinates (r, θ), the stream function and velocity potential for both the uniform flow and the doublet are given by:

$$\phi_U = V_\infty r\cos(\theta) \qquad \text{(velocity potential: uniform flow)} \tag{2.0.2}$$

$$\psi_U = V_\infty r\sin(\theta) \qquad \text{(stream function: uniform flow)} \tag{2.0.3}$$

$$\phi_D = \frac{\kappa}{2\pi}\frac{\cos(\theta)}{r} \qquad \text{(velocity potential: doublet)} \tag{2.0.4}$$

$$\psi_D = -\frac{\kappa}{2\pi}\frac{\sin(\theta)}{r} \qquad \text{(stream function: doublet)} \tag{2.0.5}$$

where V_∞ is the freestream speed and κ is the doublet strength.
Superposing the uniform flow and doublet stream functions gives

$$\begin{aligned}
\psi &= \psi_U + \psi_D && (2.0.6)\\
&= V_\infty r\sin(\theta) - \frac{\kappa}{2\pi}\frac{\sin(\theta)}{r} && (2.0.7)\\
&= V_\infty r\sin(\theta)\left(1 - \frac{R^2}{r^2}\right), && (2.0.8)
\end{aligned}$$

where the cylinder radius is defined by

$$R^2 \equiv \frac{\kappa}{2\pi V_\infty}. \tag{2.0.9}$$

The corresponding velocity field in polar coordinates follows from the stream function relations $V_r = \frac{1}{r}\frac{\partial\psi}{\partial\theta}$ and $V_\theta = -\frac{\partial\psi}{\partial r}$:

$$\begin{aligned}
V_r &= \frac{1}{r}\frac{\partial\psi}{\partial\theta} = \frac{1}{r}\left(V_\infty r\cos(\theta)\right)\left(1 - \frac{R^2}{r^2}\right) && (2.0.10)\\
&= \left(1 - \frac{R^2}{r^2}\right)V_\infty\cos(\theta), && (2.0.11)\\
V_\theta &= -\frac{\partial\psi}{\partial r} && (2.0.12)\\
&= -\left(1 + \frac{R^2}{r^2}\right)V_\infty\sin(\theta). && (2.0.13)
\end{aligned}$$

At the surface of the cylinder, $r = R$, the velocity field is:

$$\begin{aligned}
V_r &= \left(1 - \frac{R^2}{R^2}\right)V_\infty\cos(\theta) = 0, && (2.0.14)\\
V_\theta &= -\left(1 + \frac{R^2}{R^2}\right)V_\infty\sin(\theta) = -2V_\infty\sin(\theta). && (2.0.15)
\end{aligned}$$

This aligns with the assumption that the velocity normal to the stationary surface must be zero, and due to polar coordinates, the velocity normal to the surface is given by V_r. Since V_θ is perpendicular to the surface, due to construction of the coordinate system, it's equivalent to a tangential velocity, and since normal velocity is zero, $V = V_\theta = V_\infty\sin(\theta)$.
The pressure coefficient is given by:

$$\begin{aligned}
C_p &= 1 - \frac{V^2}{V_\infty^2} && (2.0.16)\\
V &= -2V_\infty\sin(\theta) && (2.0.17)\\
\frac{V^2}{V_\infty^2} &= 4\sin^2\theta && (2.0.18)
\end{aligned}$$

$$\boxed{C_p = 1 - \frac{V^2}{V_\infty^2} = 1 - 4\sin^2\theta} \tag{2.0.19}$$

3 Experimental Apparatus and Procedures

3.1 Experimental Apparatus

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3.2 Operating Technique

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4 Results

4.1 Operating Conditions

Table 4.1.1: Ambient Room Conditions during Experiment

Quantity	Value	Uncertainty
Room Temperature, T	$25\,°\mathrm{C}$	$\pm 0.5\,°\mathrm{C}$
Room Pressure, P	$1009.0\,\mathrm{hPa}$	$\pm 0.05\,\mathrm{hPa}$

The density of air (ρ) can be calculated using the Ideal Gas Law:

$$\rho = \frac{P}{RT} \tag{4.1.1}$$

where:

- P is the absolute pressure (in SI units: Pascals Pa)
- T is the absolute temperature (in Kelvin K)
- R is the specific gas constant for dry air, $R = 287.05\,\frac{\mathrm{J}}{\mathrm{kg\,K}}$

Given measurements:

$$P = 1009.0\,\mathrm{hPa} = 100\,900\,\mathrm{Pa} \tag{4.1.2}$$

$$T = 25\,\mathrm{C} = 25 + 273.15 = 298.15\,\mathrm{K} \tag{4.1.3}$$

$$R = 287.05\,\frac{\mathrm{J}}{\mathrm{kg\,K}} \tag{4.1.4}$$

$$\rho = \frac{100\,900\,\mathrm{Pa}}{287.05\,\frac{\mathrm{J}}{\mathrm{kg\,K}}} \times 298.15\,\mathrm{K} \tag{4.1.5}$$

Calculate:

$$\rho = \frac{100900}{287.05 \times 298.15} \tag{4.1.6}$$

$$= \frac{100900}{85542.5} \tag{4.1.7}$$

$$\approx 1.18\,\frac{\mathrm{kg}}{\mathrm{m}^3} \tag{4.1.8}$$

Thus, **the ambient air density during the experiment is**:

$$\boxed{\rho = 1.18\,\frac{\mathrm{kg}}{\mathrm{m}^3}} \tag{4.1.9}$$

4.2 Measurements

Table 4.2.1: Measured Relative Presssure, Pressure Coefficient, and Pressure vs. Angle of Attack

`./outputs/text/pressure_vs_theta.csv`

	θ [°]	$P - P_\infty$ [Pa]	C_P	P [Pa]
01	0.0	744.069	1.0000	101644.069
02	5.0	714.698	0.9481	101614.698
03	10.0	675.536	0.8961	101575.536
04	15.0	577.632	0.7564	101477.632
05	20.0	450.357	0.5897	101350.357
06	25.0	313.292	0.4156	101213.292
07	30.0	146.856	0.1923	101046.856
08	35.0	-39.162	-0.0519	100860.838
09	40.0	-225.179	-0.2987	100674.821
10	45.0	-411.196	-0.5455	100488.804
11	50.0	-577.632	-0.7564	100322.368
12	55.0	-675.536	-0.8846	100224.464
13	60.0	-783.230	-1.0390	100116.770
14	65.0	-822.392	-1.0909	100077.608
15	70.0	-832.182	-1.0897	100067.818
16	75.0	-763.650	-1.0000	100136.350
17	80.0	-744.069	-0.9744	100155.931
18	85.0	-724.488	-0.9610	100175.512
19	90.0	-714.698	-0.9481	100185.302
20	95.0	-695.117	-0.9221	100204.883
21	100.0	-695.117	-0.9103	100204.883
22	105.0	-704.907	-0.9351	100195.093
23	110.0	-714.698	-0.9481	100185.302
24	115.0	-714.698	-0.9481	100185.302
25	120.0	-714.698	-0.9481	100185.302
26	125.0	-734.279	-0.9740	100165.721
27	130.0	-744.069	-0.9870	100155.931
28	135.0	-744.069	-0.9870	100155.931
29	140.0	-753.859	-1.0000	100146.141
30	145.0	-763.650	-1.0130	100136.350
31	150.0	-753.859	-1.0000	100146.141
32	155.0	-744.069	-0.9870	100155.931
33	160.0	-724.488	-0.9737	100175.512
34	165.0	-724.488	-0.9610	100175.512
35	170.0	-704.907	-0.9351	100195.093
36	175.0	-704.907	-0.9351	100195.093
37	180.0	-704.907	-0.9351	100195.093

4.3 Quantities of Interest

4.3.1 Mean Wind Tunnel Velocity

The mean wind tunnel velocity, U_∞—being the same as the freestream velocity—can be obtained directly from the pitot-static pressure readings.

4.3.2 Coefficient of Pressure

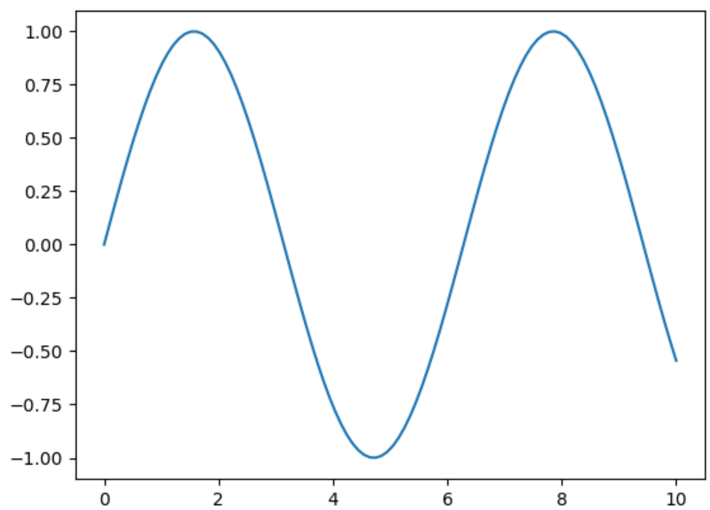

Figure 4.3.1: Pressure Coefficient C_p vs. Angle of Attack θ

4.3.3 Coefficient of Drag

The theoretically predicted drag coefficient for a cylinder in inviscid flow is zero. This result arises because, under the assumption of inviscid (frictionless) flow, the skin friction coefficient c_f is zero, and the symmetry of both the cylinder and the flow ensures that the pressure distribution is identical on the front and back surfaces. As a result, the opposing pressure forces cancel each other, leading to no net drag. This phenomenon is known as **D'Alembert's Paradox**.
Performing numerical analysis using the equations from the lecture:

$$C_d = \frac{D}{\rho_\infty} \tag{4.3.1}$$

$$D = R \int_0^{2\pi} P \cos(\theta)\, d\theta \tag{4.3.2}$$

Due to cylinder symmetry:

$$D = R \int_0^{2\pi} P \cos(\theta)\, d\theta = 2R \int_0^{\pi} P \cos(\theta)\, d\theta \tag{4.3.3}$$

To numerically calculate the integral, we used the trapezoidal approximation, and the results of C_d vs. θ are shown in the figure below:

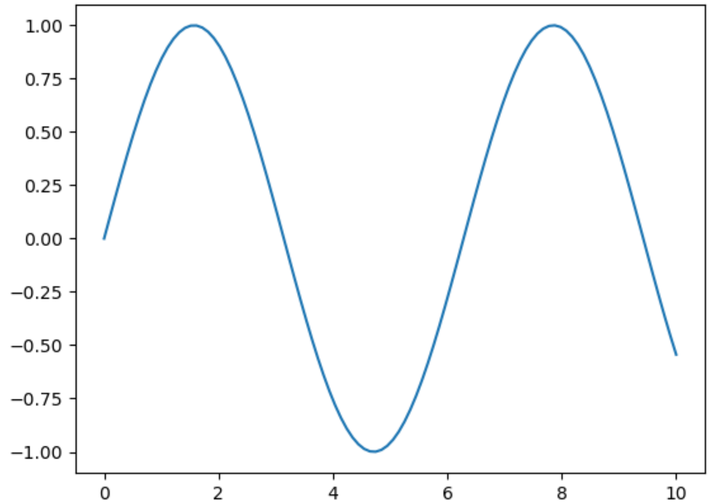

Figure 4.3.2: Drag Coefficient C_d vs. Angle of Attack θ

5 Analysis and Discussion

5.1 Data Conversion

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5.2 Interpretation

As can be seen in 4.3.1, the actual coefficient of drag is not zero. This is due to the fact that in real world air is viscous and it's coefficient of friction $c_f \neq 0$. This combined with potential flow detachment leads to a non-zero drag coefficient.
With regard to the coefficient of pressure...

6 Summary and Conclusions

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7 References

[1] J. D. Anderson and C. P. Cadou, *Fundamentals of Aerodynamics*, 7th ed. McGraw-Hill Education, March 17, 2023, pp. 209–324, ISBN: 9781266076442 (see p. 1).

8 Appendices

8.1 Data

Table 8.1.1: Raw Measurements of Pressure Taps vs. Angle of Attack

`./data/pressure_vs_theta.csv`

	θ [°]	P_∞ [cm]	P_0 [cm]	P [cm]
01	0	15.4	23.0	23.0
02	5	15.4	23.1	22.7
03	10	15.4	23.1	22.3
04	15	15.4	23.2	21.3
05	20	15.3	23.1	19.9
06	25	15.4	23.1	18.6
07	30	15.4	23.2	16.9
08	35	15.4	23.1	15.0
09	40	15.4	23.1	13.1
10	45	15.4	23.1	11.2
11	50	15.3	23.1	9.4
12	55	15.3	23.1	8.4
13	60	15.4	23.1	7.4
14	65	15.4	23.1	7.0
15	70	15.4	23.2	6.9
16	75	15.4	23.2	7.6
17	80	15.4	23.2	7.8
18	85	15.4	23.1	8.0
19	90	15.4	23.1	8.1
20	95	15.4	23.1	8.3
21	100	15.4	23.2	8.3
22	105	15.4	23.1	8.2
23	110	15.4	23.1	8.1
24	115	15.4	23.1	8.1
25	120	15.4	23.1	8.1
26	125	15.4	23.1	7.9
27	130	15.4	23.1	7.8
28	135	15.4	23.1	7.8
29	140	15.4	23.1	7.7
30	145	15.4	23.1	7.6
31	150	15.4	23.1	7.7
32	155	15.4	23.1	7.8
33	160	15.4	23.0	8.0
34	165	15.4	23.1	8.0
35	170	15.4	23.1	8.2
36	175	15.4	23.1	8.2
37	180	15.4	23.1	8.2

8.2 Code

For the sake of brevity, only the code files that are key to the analysis are included below. However, in the spirit of completeness, the repository containing the complete data, source code, and notes for this report can be found at github:vaisriv/enae464-lab01.

Listing 8.2.1: Index File
`./src/index.py`

```python
import os
import sys
import csv
import numpy as np
import matplotlib.pyplot as plt

def main():
    # ── Physical constants & ambient conditions ──────────────────────────────
    P_AMB     = 100900.0 # Ambient pressure [Pa]
    T_AMB     = 298.15   # Ambient temperature [K]
    RHO_AIR   = 1.18     # Air density [kg/m^3]
    RHO_WATER = 998.0    # Water density [kg/m^3]
    G         = 9.81     # Gravitational acceleration [m/s^2]
    RADIUS    = 0.635e-2 # Cylinder Radius [m]

    # ── File paths ───────────────────────────────────────────────────────────
    INPUT_CSV  = "./data/pressure_vs_theta.csv"
    OUTPUT_CSV = "./outputs/text/pressure_vs_theta.csv"
    OUTPUT_FILE = "./outputs/text/summary.txt"

    # ── Read CSV ─────────────────────────────────────────────────────────────
    theta_deg_list = []
    P_inf_raw_list = []
    P_0_raw_list   = []
    P_raw_list     = []

    with open(INPUT_CSV, newline="") as f:
        reader = csv.DictReader(f)
        for row in reader:
            theta_deg_list.append(float(row["theta_deg"]))
            P_inf_raw_list.append(float(row["P_inf"]))
            P_0_raw_list.append(float(row["P_0"]))
            P_raw_list.append(float(row["P"]))

    N = len(theta_deg_list)
    print(f"Read {N} data points from '{INPUT_CSV}'.")

    # ── Convert water column heights to differential pressures ──────────────
    # The manometer readings are heights (in cm of water):
    #     ΔP = ρ_water · g · Δh
    #     P_actual = P_ref − ρ_water · g · h_reading
```

```
    #     P_surface - P_freestream  =  ρ_water · g · (h_surface - h_inf)
    #     q_inf = P_0 - P_inf = ρ_water · g · (h_0 - h_inf)
    SCALE = 1e-2  # Measurements taken in cm (convert to meters)

    # ── Compute differential pressures ──────────────────────────────────
    # For each data point we have a corresponding P_inf and P_0 reading,
    # so we compute per-row to account for any drift.
    delta_P_list = [] # P_surface - P_freestream  [Pa]
    q_inf_list   = [] # dynamic pressure  [Pa]
    U_inf_list   = [] # freestream velocity [m/s]
    Cp_list      = [] # pressure coefficient

    for i in range(N):
        h_inf = P_inf_raw_list[i] * SCALE # freestream static tap height [m]
        h_0   = P_0_raw_list[i]   * SCALE # stagnation tap height [m]
        h_p   = P_raw_list[i]     * SCALE # surface tap height [m]

        # (P_surface - P_inf) in Pa
        # Higher reading = higher pressure, so:
        delta_P = RHO_WATER * G * (h_p - h_inf)

        # Dynamic pressure  q = P_total - P_static = rho_w * g * (h_0 - h_inf)
        q_inf = RHO_WATER * G * (h_0 - h_inf)

        if q_inf <= 0:
            print(f"WARNING row {i}: q_inf = {q_inf:.2f} Pa <= 0  (h_inf={h_inf:.4f}, h_0={h_0:.4f})")
            # Use absolute value as fallback but flag it
            q_inf = abs(q_inf) if abs(q_inf) > 1e-6 else 1e-6

        U_inf = (2.0 * q_inf / RHO_AIR) ** 0.5 # Bernoulli: q = 0.5 * rho * U^2

        Cp = delta_P / q_inf # Cp = (P - P_inf) / q_inf  = (P - P_inf) / (0.5 * rho * U^2)

        delta_P_list.append(delta_P)
        q_inf_list.append(q_inf)
        U_inf_list.append(U_inf)
        Cp_list.append(Cp)

    # TODO: fig: pressure coefficient vs angle of attack
    # A plot of \(C_p\) vs \(\theta\) from the experiment, plotted simultaneous (on the same graph) with the pressure coefficient that is given by inviscid theory.


    # ── Summary statistics ──────────────────────────────────────────
    q_avg   = sum(q_inf_list) / N
    U_avg   = sum(U_inf_list) / N
```

```python
    with open(OUTPUT_FILE, "w", newline="") as f:
        f.write(f"Ambient conditions:")
        f.write(f"  P_amb   = {P_AMB} Pa")
        f.write(f"  T_amb   = {T_AMB} K")
        f.write(f"  rho_air = {RHO_AIR} kg/m^3")
        f.write(f"\nDerived quantities (averages over all rows):")
        f.write(f"  q_inf   = {q_avg:.2f} Pa")
        f.write(f"  U_inf   = {U_avg:.2f} m/s")

    print(f"\nSummary written to '{OUTPUT_FILE}'.")

    # ── Write output table ─────────────────────────────────────
    with open(OUTPUT_CSV, "w", newline="") as f:
        writer = csv.writer(f)
        writer.writerow(["theta_deg", "P-P_inf_Pa", "Cp", "P_Pa"])

        for i in range(N):
            # P_actual = P_amb + (P_surface - P_inf) i.e. ambient + differential
            P_actual = P_AMB + delta_P_list[i]
            writer.writerow([
                f"{theta_deg_list[i]:.1f}",
                f"{delta_P_list[i]:.3f}",
                f"{Cp_list[i]:.4f}",
                f"{P_actual:.3f}",
            ])

    print(f"\nResults written to '{OUTPUT_CSV}'.")


if __name__ == "__main__":
    main()
```